Informatics Institute Of Technology

Affiliated With

University Of Westminster

FINAL YEAR PROJECT PROPOSAL

Adaptive Explainable Ensemble with NLP Fusion for Pre-Launch Steam Game Reception Prediction under Varying Feature Availability

Literature Review

Student details:

Name: Heshan Kaushith Ratnaweera

Degree program: Computer Science BSc Honours with Industry Placement

Student Identification: W2082289(UOW) | 20222094(IIT)

Contents

2.1 Introduction

The global video game industry has grown into a major entertainment sector, with estimates ranging from \$152 billion to \$300 billion depending on the year and revenue streams considered (Ilyanov et al., 2020; Peri, 2024; Fan, 2023). On the Steam platform, user review scores serve as a key indicator of player reception and commercial success (Kraemer et al., 2025; Santos et al., 2019). However, accurately predicting whether a game will achieve positive user reception (≥75% positive reviews) using only pre-launch metadata remains a significant challenge for developers.

This literature review synthesizes research on the reliability of Steam user reviews, machine learning approaches to game success prediction, and methods from cross-domain pre-launch prediction studies. Its purpose is to identify established techniques, evaluate their strengths and limitations, and highlight gaps that justify an adaptive explainable ensemble model incorporating NLP fusion.

Key themes explored include: reliability of user review scores as quality signals, tabular ensemble classifiers for binary prediction tasks, hybrid models combining structured features with pre-trained language model embeddings, techniques for addressing class imbalance, adaptive routing mechanisms, and post-hoc interpretability methods. The review draws primarily on studies published between 2020 and 2026, with foundational works included only where essential.

The chapter is structured as follows: Section 2.2 outlines background concepts and fundamental theories. Section 2.3 presents a thematic review of the literature. Section 2.4 offers a critical analysis and synthesis. Section 2.5 identifies research gaps, and Section 2.6 provides a chapter summary.

2.2 Background Concepts / Fundamental Theories

This section defines the core concepts underpinning the research. A clear understanding of these terms is essential before examining the literature.

Pre-launch Metadata

Pre-launch metadata refers to information available about a video game before its public release on platforms such as Steam. This typically includes structured attributes such as genre, price, DLC (downloadable content) count, age rating, content tags, supported platforms, and developer-written textual descriptions (short_description). These features are determined during the design and production phases and do not rely on post-release player behaviour or review data (Aleem and Noor, 2024; Sahu et al., 2023).

User Reception and Binary Classification

User reception on Steam is commonly measured through review scores, with positive reception often defined as 75% or higher positive reviews. For research in this domain, the prediction task is naturally formulated as a binary classification problem: determining whether a game will achieve “well-received” or “not well-received” status based solely on pre-launch features. This binary framing aligns with analogous success prediction tasks across domains (Kraemer et al., 2025; Sahu et al., 2023).

Tabular Data and Ensemble Methods

Tabular data consists of structured rows and columns, typical of game metadata. Tree-based ensemble methods such as XGBoost, Random Forest, and Gradient Boosting are widely recognised as strong performers on tabular binary classification tasks, often outperforming linear models and deep learning approaches on typical benchmarks (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024).

Hybrid NLP-Tabular Fusion
Hybrid models combine structured tabular features with textual information processed through pre-trained language models. BERT (Bidirectional Encoder Representations from Transformers; Devlin et al., 2019) introduced the transformer-based pre-training paradigm that underpins modern NLP embeddings. Sentence-BERT (Reimers and Gurevych, 2019) extends this to produce fixed-length sentence embeddings that can be fused with tabular data via concatenation or more advanced mechanisms. Such fusion has shown performance gains in domains with both structured and textual pre-launch information (Gunduz, 2024; Kasneci and Kasneci, 2024; Gameiro, 2025; Qin et al., 2023).

Class Imbalance
Class imbalance occurs when one class (e.g., well-received games) significantly outnumbers the other. This is common in reception prediction tasks. Techniques to address it include oversampling via SMOTE (Synthetic Minority Oversampling Technique; Chawla et al., 2002), class weighting, and ensemble methods (Abdelhamid and Desai, 2024; Carvalho et al., 2025).

Post-hoc Explainability (SHAP and LIME)
Explainability methods help interpret black-box model predictions. SHAP (SHapley Additive exPlanations) provides consistent feature attribution based on game theory (Lundberg and Lee, 2017), while LIME (Local Interpretable Model-agnostic Explanations) fits a simple surrogate model locally around individual predictions (Ribeiro et al., 2016). Both are model-agnostic and commonly used for tabular and hybrid models (Yasui and Sato, 2025; Salih et al., 2023).

Adaptive Architectures and Confidence-based Routing
Adaptive systems route predictions to different specialist models depending on available features or prediction confidence. Confidence is typically measured using the maximum predicted probability from a model's output (Alajmi and Pergola, 2025/2026; Wang et al., 2024). This enables graceful handling of varying feature availability during game development.

Feature Engineering and Derived Features
Feature engineering involves creating new variables from raw data, such as interaction terms (e.g., price-to-DLC ratio) or domain-specific indices. This process often yields substantial performance improvements in structured prediction tasks (Sahu et al., 2023; Tschalzev et al., 2024).

These concepts collectively underpin the literature examined in the sections that follow.

2.3 Thematic Literature Review

This section organizes the literature into key themes relevant to the development of an adaptive explainable ensemble model for pre-launch Steam game reception prediction.

2.3.1 Reliability of Steam User Reviews as Quality Signals

Steam user review scores are frequently used as proxies for game quality, yet research indicates they reflect player-perceived quality only moderately. Kraemer et al. (2025) highlight that higher user and critic review valence tends to predict higher sales, but the effect size varies by platform, review type, and game segment. Santos et al. (2019) analyzed over one million Metacritic reviews and found user scores to be more polarized and vulnerable to manipulation compared to critic scores.

Busurkina et al. (2020) similarly note inconsistencies between user reviews and expert assessments across genres, finding that review sentiment varies substantially by game type and reviewer population. Li et al. (2021) emphasize that reviews are strongest at capturing playability and concrete flaws rather than serving as a universal quality metric. These findings underscore the commercial importance of predicting reception early while revealing limitations in using post-launch reviews for model training labels.

Collectively, these findings caution against treating Steam review scores as unmediated ground truth, and support the use of a ≥75% positive review threshold as a conservative and widely-used operationalization of binary reception quality in studies of this kind (Santos et al., 2019; Kraemer et al., 2025).

2.3.2 Tabular Ensemble Methods for Binary Classification

Tree-based ensemble methods dominate performance on tabular binary classification tasks. Shwartz-Ziv and Armon (2021), Grinsztajn et al. (2022), and Uddin and Lu (2024) consistently demonstrate that XGBoost (Chen and Guestrin, 2016), Random Forest (Breiman, 2001), and gradient boosting algorithms outperform linear models, SVMs, k-NN, and many deep learning approaches across broad tabular benchmarks. Bentéjac et al. (2019) provide a comparative analysis of gradient boosting variants, confirming their robustness across diverse classification tasks.

However, advantages are not universal. Karlsson et al. (2024) note that some complex datasets favor newer deep models, highlighting the value of adaptive model selection. These studies provide strong justification for applying specialist tree ensemble classifiers across graduated feature tiers in tabular game-data classification.

While the weight of evidence strongly endorses tree ensembles as a backbone classifier for tabular data, the dataset-dependent caveat raised by Karlsson et al. (2024) underscores that no single model should be assumed optimal across all feature configurations, reinforcing the case for adaptive, multi-tier model selection in any classification system operating over heterogeneous tabular inputs.

2.3.3 Pre-launch Prediction in Cross-Domain Studies

Several domains have addressed structurally similar pre-launch prediction problems. In film, Sahu et al. (2023) achieved high accuracy using derived features and hybrid deep ensembles on pre-release metadata. Aleem and Noor (2024) demonstrated the effectiveness of RF, SVM, XGBoost, and KNN for Android app success prediction using pre-launch attributes (category, price, rating), offering a close parallel to Steam game metadata.

In crowdfunding, Gunduz (2024) showed that BERT representations of creator-written text outperform traditional embeddings for campaign success prediction, with precision gains of over 10% compared to

TF-IDF baselines. Alajmi and Pergola (2025/2026) demonstrated that confidence-based routing between specialist classifiers improves robustness in multi-class NLP tasks, motivating its potential transfer to gaming-domain pre-launch prediction problems.

Critically, these cross-domain studies share a common methodological pattern: they each isolate pre-launch features, apply derived or engineered variables, and evaluate binary or categorical success outcomes. This structural similarity to the Steam reception prediction problem strongly validates the transfer of their component techniques to the gaming context. However, no reviewed study combines tiered specialist models, derived features, NLP fusion, and confidence-based routing within a single unified adaptive framework, representing a substantive and unexplored methodological gap in the gaming domain.

2.3.4 Hybrid Models: Integrating NLP with Structured Tabular Features

Hybrid approaches consistently improve performance when textual and structured data are available. Qin et al. (2023) reported nearly 24% improvement over single-modality baselines by fusing transformer-based text features with structured metadata in movie rating prediction. Kasneci and Kasneci (2024) found that RoBERTa and GPT-2 embeddings concatenated with tabular features notably boost XGBoost and CatBoost on imbalanced datasets.

Gameiro (2025) introduced a synergistic fusion layer with Sentence-BERT embeddings and structured features for lyrical classification, achieving superior Macro F1 scores. Maarouf et al. (2024) further demonstrated effective SHAP application across fused LLM embeddings and financial features, confirming that post-hoc explainability integrates well with hybrid architectures. Bhavesh Kumar and Pande (2023) extended SHAP-based attribution specifically to hybrid models combining text and tabular inputs, showing that individual feature contributions remain interpretable even when embedding dimensions are high-dimensional. These results collectively suggest that short-form developer-written text such as the Steam short_description field represents a viable and high-signal modality for hybrid NLP-tabular fusion, particularly when paired with post-hoc explainability methods capable of attributing predictions across both structured and embedded features.

A key limitation across most of these hybrid studies is their reliance on simple feature concatenation as the sole fusion strategy, with limited rigorous comparison of alternatives such as gated attention or cross-modal projection. This suggests that the choice of fusion mechanism warrants empirical scrutiny and should not be treated as a settled design decision.

2.3.5 Class Imbalance Handling and Adaptive Ensemble Techniques

Class imbalance is prevalent in reception prediction. Abdelhamid and Desai (2024) and Carvalho et al. (2025) review resampling, class weighting, and hybrid strategies, finding no universally superior method but noting SMOTE’s utility alongside weighting.

Stacking ensembles and meta-learners improve multi-model performance by combining the probability outputs of multiple base classifiers through a secondary learner (Liao et al., 2022; Şencan et al., 2026). This approach offers a principled mechanism for aggregating predictions from multiple specialist classifiers, with demonstrated performance benefits in analogous prediction tasks across domains.

The absence of a universally superior imbalance-handling strategy across these studies suggests that empirical comparison across multiple candidate techniques—including resampling, class weighting, and no-correction baselines—is preferable to prescribing a single approach without domain-specific validation.

2.3.6 Adaptive ML and Confidence-Based Routing

Confidence-based routing, where inputs are directed to the specialist model with the highest predicted probability (predict_proba), has shown promise in NLP and healthcare domains (Alajmi and Pergola, 2025/2026; Wang et al., 2024). This approach provides a flexible alternative to rigid rule-based routing under varying feature availability, allowing the system to dynamically select the most confident eligible model rather than following a fixed decision tree. Le Morvan et al. (2021) further demonstrate that adaptive architectures designed for missing data outperform imputation-based baselines, providing additional justification for confidence-gated routing as a robust design principle in systems that must operate under varying input conditions.

Its application to game metadata prediction, however, remains untested; the degree to which confidence scores derived from tabular and hybrid classifiers translate robustly across structurally different feature tiers is an open empirical question that the literature has yet to examine.

2.3.7 Post-hoc Explainability Methods

SHAP and LIME provide valuable interpretability for black-box models. Lundberg and Lee (2017) introduced SHAP as a unified framework for feature attribution, while Ribeiro et al. (2016) proposed LIME for local surrogate explanations. Recent work by Yasui and Sato (2025) and Salih et al. (2023) extends these for improved fidelity in tabular and hybrid settings. Dual SHAP+LIME usage enables both global and instance-level insights, critical for developer-facing reception risk assessment.

While SHAP and LIME are individually well-validated, their joint application to hybrid NLP-tabular architectures remains relatively underexplored, and reconciling global attribution from SHAP with local surrogate explanations from LIME in high-dimensional embedding spaces represents an unresolved methodological challenge that warrants explicit treatment in any system combining structured tabular and textual modalities.

2.4 Critical Analysis and Synthesis

This section critically evaluates the literature by identifying overarching trends, areas of agreement, contradictions, and recurring limitations. The analysis draws together insights from the thematic review to highlight methodological strengths and weaknesses relevant to pre-launch Steam game reception prediction.

Trends

A clear trend across recent studies (2020–2026) is the dominance of tree-based ensemble methods on tabular data. Multiple sources confirm that XGBoost, Random Forest, and gradient boosting consistently rank among the strongest classifiers for binary tasks, outperforming linear models and many deep learning approaches (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024; Bentéjac et al., 2019). Another prominent trend is the growing use of hybrid architectures that fuse pre-trained language model embeddings with structured features, particularly in domains involving creator-written text (Qin et al., 2023; Kasneci and Kasneci, 2024; Gameiro, 2025; Gunduz, 2024). There is also increasing attention to explainability, with SHAP and LIME frequently applied to provide actionable insights (Lundberg and Lee, 2017; Ribeiro et al., 2016; Yasui and Sato, 2025).

In cross-domain pre-launch prediction (film, app stores, crowdfunding), derived feature engineering and adaptive techniques are emerging as high-impact practices (Sahu et al., 2023; Aleem and Noor, 2024; Alajmi and Pergola, 2025/2026).

Agreements

The literature shows strong consensus on several points. First, tree ensembles generally deliver robust performance on structured tabular data, especially under class imbalance (Uddin and Lu, 2024; Grinsztajn et al., 2022). Second, hybrid NLP-tabular models outperform single-modality baselines when semantically rich text (such as descriptions or blurbs) is available (Qin et al., 2023; Gameiro, 2025; Maarouf et al., 2024). Third, pre-launch metadata and developer-written text contain meaningful predictive signals for success or reception across domains (Sahu et al., 2023; Gunduz, 2024; Aleem and Noor, 2024). Finally, post-hoc explainability tools like SHAP and LIME are widely accepted as effective for building trust in black-box predictions (Salih et al., 2023; Yasui and Sato, 2025).

Current Challenges, Contradictions, and Limitations

Despite these agreements, important contradictions and limitations exist. While most studies praise tree ensembles, Karlsson et al. (2024) demonstrate that advantages are not universally large and that complex datasets may favour newer deep models. Similarly, the superiority of specific imbalance-handling techniques remains dataset-dependent, with no single method dominating across all settings (Abdelhamid and Desai, 2024; Carvalho et al., 2025).

A major limitation across gaming-domain studies is heavy reliance on post-launch features (e.g., review counts, player activity, peak CCU), rendering them unsuitable for true pre-launch prediction (as seen in multiple domain papers such as Ma et al., 2025; Pramanik, 2025; Rahman et al., 2024). Most gaming research is descriptive or reactive rather than predictive (Santos et al., 2019; Busurkina et al., 2020). Additionally, few studies address varying feature availability during development, and rule-based routing is rarely compared against confidence-based alternatives (Alajmi and Pergola, 2025/2026; Wang et al., 2024).

User review scores, while commercially relevant (Kraemer et al., 2025), suffer from polarization, manipulation risks, and genre biases (Santos et al., 2019; Coronado-Bl'azquez, 2024), limiting their reliability as ground truth labels. Many hybrid studies also lack rigorous evaluation of fusion strategies beyond simple concatenation.

Synthesis

Overall, the literature strongly supports the use of adaptive tree ensembles, hybrid NLP fusion via Sentence-BERT, derived features, confidence-based routing, and dual SHAP+LIME explainability. However, these techniques have rarely been combined in a single system tailored to the progressive nature of game development metadata availability. This synthesis reveals both mature methodological building blocks and clear opportunities for integration that remain unrealized in existing work.

2.5 Research Gap Analysis

The reviewed literature reveals several important gaps that collectively justify the need for the proposed adaptive explainable ensemble with NLP fusion for pre-launch Steam game reception prediction.

Practical Gap
Despite the commercial importance of user reception on Steam (Kraemer et al., 2025; Ilyanov et al., 2020; Peri, 2024), existing studies in the gaming domain predominantly rely on post-launch metrics such as review counts, peak concurrent users, player activity, and engagement data (Ma et al., 2025; Pramanik, 2025; Rahman et al., 2024; Hu et al., 2024; Teja et al., 2023). No reviewed work in the gaming domain formulates the problem as a true pre-launch binary classification task using only metadata and developer-written descriptions available before release. Developers therefore lack evidence-based tools to assess reception risk during design, pricing, and production phases when interventions are most feasible.

Methodological Gap
While tree-based ensembles perform strongly on tabular data (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024) and hybrid NLP-tabular models show promise in other domains (Qin et al., 2023; Kasneci and Kasneci, 2024; Gameiro, 2025; Gunduz, 2024), these approaches have not been systematically combined or adapted for the gaming context. Most gaming studies do not explore derived feature engineering (Sahu et al., 2023), confidence-based model routing (Alajmi and Pergola, 2025/2026; Wang et al., 2024), or specialist models trained on graduated feature tiers that reflect progressive metadata availability during development.

Evaluation and Data Gap
There is a notable absence of labelled benchmarks that pair pre-launch Steam metadata with binary reception outcomes (≥75% positive reviews). Existing game-related research tends to focus on regression, popularity, revenue, or churn rather than binary reception quality prediction (Pramanik, 2025; Rahman et al., 2024; Mulla et al., 2025). Furthermore, few studies address class imbalance handling specifically within pre-launch reception prediction (Abdelhamid and Desai, 2024; Carvalho et al., 2025), and explainability is rarely integrated into adaptive multi-model systems for this domain.

Deployment Gap
Current systems assume a complete, fixed feature set at prediction time. None explicitly support varying feature availability across development stages (concept → production → pre-release), nor do they combine confidence-based routing with dual SHAP and LIME explainability for developer decision support (Yasui and Sato, 2025; Salih et al., 2023; Alajmi and Pergola, 2025/2026).

Description Gap
No reviewed study in the gaming domain has examined developer-written text (short_description) as a predictive feature for game reception. In adjacent domains, creator-written descriptions have demonstrated strong predictive value for campaign and product success when processed through transformer-based models (Gunduz, 2024; Kasneci and Kasneci, 2024; Gameiro, 2025). The absence of text-based modelling in gaming reception research represents a significant gap, given the widespread availability of developer-written descriptions on Steam and the demonstrated predictive value of such text in adjacent domains.

Routing Gap

Existing gaming-domain prediction systems use either a single fixed model or rigid rule-based selection between models. No reviewed study employs confidence-based routing, whereby the system dynamically selects the most confident eligible specialist model at prediction time using predicted probability (Alajmi and Pergola, 2025/2026; Wang et al., 2024). This architectural gap is significant: confidence-based routing directly enables graceful degradation under partial feature availability, which is a core requirement when predicting reception at different stages of game development (concept, production, pre-release).

Synthesis of Gaps

Cross-domain studies demonstrate the viability of pre-launch prediction, hybrid NLP-tabular fusion, derived feature engineering, and adaptive routing, yet these insights remain unapplied to Steam game reception within a unified and explainable framework. Addressing this gap would require a system that combines adaptive specialist ensembles operating across graduated feature tiers, developer-written text as a predictive modality, confidence-based routing, and integrated post-hoc explainability—a combination that no reviewed study currently provides.

2.6 Chapter Summary

This chapter has reviewed the current state of research relevant to pre-launch prediction of Steam game user reception. It began by establishing fundamental concepts such as pre-launch metadata, binary reception classification, tabular ensembles, hybrid NLP-tabular fusion, class imbalance handling, and post-hoc explainability techniques. The thematic review then examined key areas including the reliability of Steam user reviews, the strength of tree-based ensemble methods, cross-domain pre-launch prediction approaches, hybrid models, adaptive techniques, and interpretability methods.

The literature demonstrates several consistent findings. Tree-based ensembles such as XGBoost, Random Forest, and gradient boosting remain highly effective for tabular binary classification tasks (Shwartz-Ziv and Armon, 2021; Grinsztajn et al., 2022; Uddin and Lu, 2024). Hybrid architectures that integrate pre-trained language model embeddings (particularly Sentence-BERT) with structured features consistently outperform single-modality baselines in analogous domains (Qin et al., 2023; Kasneci and Kasneci, 2024; Gameiro, 2025; Gunduz, 2024). Derived feature engineering, confidence-based routing, stacking ensembles, and dual SHAP+LIME explainability have also shown value across multiple studies (Sahu et al., 2023; Alajmi and Pergola, 2025/2026; Liao et al., 2022; Yasui and Sato, 2025).

However, critical limitations persist. Gaming-domain research is predominantly post-launch focused, relying on metrics unavailable during development (Ma et al., 2025; Pramanik, 2025; Rahman et al., 2024). Few studies address varying feature availability across development stages, and there is a notable absence of integrated adaptive, hybrid, and explainable systems tailored to Steam game reception prediction. User review scores, while commercially relevant (Kraemer et al., 2025), exhibit polarization, manipulation risks, and genre biases that complicate their use as prediction targets (Santos et al., 2019).

These gaps collectively define a practical and methodological opportunity: developers currently lack evidence-based tools for pre-launch reception risk assessment, and the techniques required to build such a tool—adaptive tiered ensembles, hybrid NLP-tabular fusion, confidence-based routing, derived feature engineering, and dual explainability—each have strong individual support in the literature but have not been unified in the gaming context. Chapter 3 operationalizes the approach derived from these findings.

References

Abdelhamid, M. and Desai, A. (2024) ‘Balancing the Scales: A Comprehensive Study on Tackling Class Imbalance in Binary Classification’, arXiv preprint arXiv:2409.19751. doi: 10.48550/arxiv.2409.19751.

Alajmi, A. and Pergola, G. (2025/2026) ‘When in doubt, consult: expert debate for sexism detection via confidence-based routing’, arXiv preprint arXiv:2512.23732.

Aleem, M. and Noor, N.M. (2024) ‘Predicting Android app success before Google Play Store launch using machine learning’, Library Progress International, 44(3), pp. 1907–1918. doi: 10.48165/bapas.2024.44.2.1. Available at: https://bpasjournals.com/libraryscience/index.php/journal/article/view/641.

Bentéjac, C., Csörgo, A. and Martínez-Muñoz, G. (2019) ‘A comparative analysis of gradient boosting algorithms’, Artificial Intelligence Review, 54, pp. 1937–1967. doi: 10.1007/s10462-020-09896-5.

Blagus, R. and Lusa, L. (2013) ‘SMOTE for high-dimensional class-imbalanced data’, BMC Bioinformatics, 14, p. 106. doi: 10.1186/1471-2105-14-106.

Busurkina, I., Karpenko, V., Tulubenskaya, E. and Bulygin, D. (2020) ‘Game Experience Evaluation. A Study of Game Reviews on the Steam Platform’, in Proceedings, pp. 117–127. doi: 10.1007/978-3-030-65218-0_9.

Bhavesh Kumar, S. and Pande, S.D. (2023) ‘Explainable neural network analysis on movie success prediction’, EAI Endorsed Transactions on Scalable Information Systems, 11(4). doi: 10.4108/eetsis.4435. Available at:https://doi.org/10.4108/eetsis.4435.

Breiman, L. (2001) ‘Random Forests’, Machine Learning, 45(1), pp. 5–32. doi: 10.1023/A:1010933404324.

Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P. (2002) ‘SMOTE: Synthetic Minority Over-sampling Technique’, Journal of Artificial Intelligence Research, 16, pp. 321–357. doi: 10.1613/jair.953.

Chen, T. and Guestrin, C. (2016) ‘XGBoost: A Scalable Tree Boosting System’, in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. doi: 10.1145/2939672.2939785.

Devlin, J., Chang, M.-W., Lee, K. and Toutanova, K. (2019) ‘BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding’, in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT). doi: 10.18653/v1/N19-1423.

Carvalho, M., Pinho, A. J. and Brás, S. (2025) ‘Resampling approaches to handle class imbalance: a review from a data perspective’, Journal of Big Data, 12. doi: 10.1186/s40537-025-01119-4.

Coronado-Bl'azquez, J. (2024) ‘A NLP Approach to “Review Bombing” in Metacritic PC Videogames User Ratings’, arXiv preprint arXiv:2405.06306. doi: 10.48550/arxiv.2405.06306.

Fan, N. (2023) ‘Evaluation of the Performance and Behavior of the Video Game Industry’, Advances in Economics, Management and Political Sciences. doi: 10.54254/2754-1169/30/20231425.

Gameiro, M.A. (2025) ‘Synergistic feature fusion for latent lyrical classification: a gated deep learning approach’, arXiv preprint arXiv:2511.11673.

Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022) 'Why do tree-based models still outperform deep learning on tabular data?', Advances in Neural Information Processing Systems (NeurIPS), 35, pp. 507–520. Available at: https://arxiv.org/abs/2207.08815

Gunduz, H. (2024) ‘Predicting Kickstarter success from pre-launch campaign text using BERT and traditional ML’, PeerJ Computer Science. doi: 10.7717/peerj-cs.2316.

Hu, W., Wang, Y. and Xia, R. (2024) 'Machine learning-based Steam platform game's popularity analysis', in Proceedings of the 1st International Conference on Data Science and Engineering (DSAE 2024). SCITEPRESS-Science and Technology Publications, pp. 582–588. doi: 10.5220/0012853800004547

Ilyanov, D. S., Chernysheva, T. C. and Yurevich, M. (2020) ‘Sources of economic growth in the XXI century: video game industry’, Теоретическая и прикладная экономика. doi: 10.25136/2409-8647.2020.3.31693.

Karlsson, A., Wang, T., Nowaczyk, S., Pashami, S. and Asadi, S. (2024) ‘Mind the Data, Measuring the Performance Gap Between Tree Ensembles and Deep Learning on Tabular Data’, pp. 65–76. doi: 10.1007/978-3-031-58547-0_6.

Kasneci, E. and Kasneci, G. (2024) ‘Enriching tabular data with contextual LLM embeddings: a comprehensive ablation study for ensemble classifiers’, arXiv preprint arXiv:2411.01645.

Available at: https://arxiv.org/abs/2411.01645.

Kraemer, T., Weiger, W. H. and Heidenreich, S. (2025) ‘Do all stars shine the same? Investigating the nonlinear effects of user and critic reviews on video game sales’, Journal of Business Research. doi: 10.1016/j.jbusres.2024.115034.

Li, X., Zhang, Z. and Stefanidis, K. (2021) ‘A Data-Driven Approach for Video Game Playability Analysis Based on Players’ Reviews’, Information, 12(3), p. 129. doi: 10.3390/info12030129.

Liao, Y., Peng, Y., Shi, S., Shi, V. and Yu, X. (2022) ‘Early box office prediction in China’s film market based on a stacking fusion model’, Annals of Operations Research, 308(1), pp. 321–338. doi: 10.1007/s10479-020-03804-4.

Lundberg, S. and Lee, S. (2017) 'A unified approach to interpreting model predictions', in Advances in Neural Information Processing Systems (NeurIPS), 30, pp. 4765–4774. Available at: https://arxiv.org/abs/1705.07874

Ma, J. (2025) 'Unveiling success drivers in gaming: a machine learning study across Steam, Twitch, and Metacritic', International Journal of Computer Games Technology, 2025, Article 5776632. doi: 10.1155/ijcg/5776632

Maarouf, A., Feuerriegel, S. and Pröllochs, N. (2024) ‘Predicting startup success by fusing LLM-encoded text descriptions with structured features’, arXiv preprint arXiv:2409.03668.

Mulla, R., Potharaju, S., Tambe, S.N., Joshi, S., Kale, K., Bandishti, P. and Patre, R. (2025) 'Predicting player churn in the gaming industry: a machine learning framework for enhanced retention strategies', Journal of Current Science and Technology, 15(2), Article 103. doi: 10.59796/jcst.V15N2.2025.103

Müller, D., Soto-Rey, I. and Kramer, F. (2022) 'An Analysis on Ensemble Learning Optimized Medical Image Classification With Deep Convolutional Neural Networks', IEEE Access, 10, pp. 66467–66480. doi: 10.1109/access.2022.3182399.

Peri, J. (2024) 'Video gaming industry in the US', International Journal of Science and Research Archive. doi: 10.30574/ijsra.2024.11.1.0194.

Ellington, C., Pramanik, P. and Robinson, H.K. (2025) 'MetaScoreLens: evaluating user feedback across digital entertainment systems', arXiv preprint arXiv:2601.11523. Available at: https://arxiv.org/abs/2601.11523

Qin, M., Zhou, Q., Chen, W. and Zhao, L. (2023) 'MAMRP: multi-modal data aware movie rating prediction', in Advanced Data Mining and Applications, Springer LNCS. doi: 10.1007/978-3-031 46664-9_44.

Rahman et al. (2024) 'Predict number of Steam game players', International Journal of Advanced Computer Science and Applications, 15.

Reimers, N. and Gurevych, I. (2019) 'Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks', arXiv preprint arXiv:1908.10084. doi: 10.18653/v1/d19-1410.

Ribeiro, M. T., Singh, S. and Guestrin, C. (2016) '"Why Should I Trust You?": Explaining the Predictions of Any Classifier', in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. doi: 10.1145/2939672.2939778.

Sahu, S., Kumar, R., Long, H.V. et al. (2023) 'Early-production stage prediction of movies success using K fold hybrid deep ensemble learning model', Multimedia Tools and Applications, 82, pp. 4031-4061. doi: 10.1007/s11042-022-13448-0.

Salih, A. M. A. et al. (2023) 'A Perspective on Explainable Artificial Intelligence Methods: SHAP and LIME', Advanced Intelligent Systems, 7. doi: 10.1002/aisy.202400304.

Santos, T., Lemmerich, F., Strohmaier, M. and Helic, D. (2019). What's in a Review. Proceedings of the ACM on Human-Computer Interaction, 3, pp. 1 - 22. doi: 10.1145/3359242.

Shwartz-Ziv, R. and Armon, A. (2021) 'Tabular Data: Deep Learning is Not All You Need', Information Fusion, 81, pp. 84–90. doi: 10.1016/j.inffus.2021.11.011.

Teja, A.S., Hanafi, M.L.I. and Qomariyah, N.N. (2023) 'Predicting Steam games rating with regression', in E3S Web of Conferences, 388, Article 02001. EDP Sciences. doi: 10.1051/e3sconf/202338802001

Tschalzev, A., Marton, S., Ludtke, S., Bartelt, C. and Stuckenschmidt, H. (2024). A DataCentric Perspective on Evaluating Machine Learning Models for Tabular Data. ArXiv, abs/2407.02112. doi: 10.48550/arxiv.2407.02112.

Uddin, S. and Lu, H. (2024) 'Confirming the statistically significant superiority of tree-based machine learning algorithms over their counterparts for tabular data', PLOS ONE, 19. doi: 10.1371/journal.pone.0301541.

Wang, J., Ahn, S., Dalal, T., Zhang, X., Pan, W., Zhang, Q., Chen, B., Dodge, H.H., Wang, F. and Zhou, J. (2024) 'Augmented risk prediction for Alzheimer's disease from electronic health records with large language models', arXiv preprint arXiv:2405.16413. Available at:

https://arxiv.org/abs/2405.16413.

Yasui, D. and Sato, H. (2025) 'Improving Local Fidelity and Interpretability of LIME by Replacing Only the Sampling Process With CVAE', IEEE Access, 13, pp. 53084–53099. doi: 10.1109/access.2025.3553505.

Şencan, Ö.A., Atacak, İ., Doğru, İ.A., Toklu, S., Barışçı, N. and Kılıç, K. (2026) 'FUSEPOP: A multi-modal fusion with mutual information weighting and stacked ensemble for social media popularity prediction', Applied Sciences, 16(9), 4160. doi: 10.3390/app16094160.